

Omega and Domino Pi boards, which cost around US\$ 10, have integrated Wi-Fi stacks and can run embedded Linux."

With an increasing requirement for ultra low power and data reliability, Bhaskar feels, ferroelectric random access memory (FRAM) is slowly being seen as a viable replacement for flash. FRAM offers a huge number of read/write cycles, with lesser read/write times and much lower power consumption as opposed to flash. He says, "These kinds of devices are useful in particular for applications that have very stringent requirements on power consumption."

He adds, "TI also features a whole host of development kits, both general-purpose as well as application-specific, based on our FRAM MCUs for users to evaluate."

Developers should also look for development boards integrated with onboard features for different wireless technologies like Wi-Fi, Bluetooth, Global System for Mobile communication (GSM), General Packet Radio Service (GPRS), ZigBee and near field communication (NFC) with a low-power MCU and display, opines Shah.

He says, "The ability to interface different low-power technology sensors like accelerometers, gyroscopes, global positioning systems (GPSes), temperature and humidity sensors using add-ons should be made available onboard in future."

### Boards in the spotlight and some upcoming ones

A new US\$ 9 board called Chip has recently received more than US\$ 2 million in online crowd-funding, informs Anwaarullah. He says, "This is a power-packed board that comes with onboard Wi-Fi, Bluetooth and an ARM processor. At such a low price, building powerful connected devices and scalable solutions becomes a lot easier." This board, which starts

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shipping by 2016, will create endless possibilities.

MediaTek LinkIt ONE development platform is one among the best, feels Virupaksha. He says, "It enables you to design and prototype wearables and IoT devices, using hardware and application programming interfaces (API) that are similar to those offered for Arduino boards." He adds, "It is a power pack in terms of its features as it offers Bluetooth, GSM and GPS on a single platform and, most importantly, in a very small form factor."

Raspi recently sold two million boards and is a very good board for learning basic computer programming and interfacing with hardware. Mehta says, "Apart from Raspi, we are seeing high traction for Wi-Fi boards with ESP8266 or AR9331 chipsets."

"mbed has been making quite some noise with its Cortex M0 and Cortex M3 based NXP MCU development boards supporting cloud based firmware development," says Shah. He adds, "Launchpad series by TI is quite popular among developers working on low-power battery-operated devices based on MSP430."

BLE sensor tag based on CC2541 has garnered a lot of attention ever since it was released. It has won EE Times/EDN Ace award for the best development kit and was also announced as the first Bluetooth Smart accessory to support Android (July 2013). Bhaskar says, "This design kit, while being extremely low-cost (about US\$ 25), has been extremely popular with users over the years for IoT based solution development."

Similarly, CC3200 Launchpad has been of great interest among users

looking to add Wi-Fi based connectivity to their embedded applications. Bhaskar says, "CC3200 has been an award-winning chipset, receiving several accolades in 2014 such as Electronics Products Product of the Year, Zinnov Innovation Award and Electronic Design Best Communication Product of the Year."

Arduino Zero definitely provides a huge boost to the community and provides a powerful MCU with Arduino IDE support, which makes it merrier to implement some complex projects, informs Virupaksha. He says, "Many crowd-funded boards using Nordic Semiconductor's nRF51822 have helped many BLE developers to demonstrate use cases with mobile phones and other IoT applications."

Keeping in mind the requirements of building scalable IoT applications, Anwaarullah says, "Our team has built an Arduino-compatible Wi-Fi IoT development board called Refflion for building end-to-end IoT solutions through cloud gateway and mobile applications." Made in India, Refflion can also be used by educational organisations and universities to introduce students to build connected robotics, wearables and other IoT solutions. He says, "Refflion will soon be launched internationally on crowd funding, and backers would be able to get the board and shields at a special price."

Knewron's Quickly is an IoT-ready development board, which has all the essentials required to develop IoT-related products. Anand says, "Alpha version of the board is already available for sampling, and beta version is coming out in July. This was also featured in EFY Expo 2015 held at New Delhi." ●